```
!pip install sentence-transformers
```

```
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.1.1)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from sentence-trans
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transform
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.10
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.0.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.6.1
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.16.3)
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transfor
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (4.67.3)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->
Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->s
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sente
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-tr
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transfo
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tran
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tr
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformer
Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sente
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch>=1.1
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=1
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=1.1
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /usr/local/lib/python3.12/dist-packages (from torch>=1.1
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /usr/local/lib/python3.12/dist-packages (from torch>=1.1
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->se
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->s
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tran
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.4
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers<6
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.41.0->se
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4.
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->sentence-trans
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->sentenc
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfac
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0-
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0->sen
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.20.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.20.
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hu
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
```

```
import pandas as pd

# Load dataset
df = pd.read_csv("/content/Aicte Intership Dataset.csv")

# Clean data
df = df.fillna('').astype(str)
df.columns = df.columns.str.strip().str.lower()

# Combine important fields
df["combined_text"] = (
    df["job title"] + " " +
    df["company name"] + " " +
    df["cities"] + " " +
    df["states"] + " " +
    df["stipend"] + " " +
    df["duration"]
)
```

```
from google.colab import files

uploaded = files.upload()
```

Choose files kumar gk Resume.pdf
kumar gk Resume.pdf(application/pdf) - 42510 bytes, last modified: 01/02/2026 - 100% done
Saving kumar gk Resume.pdf to kumar gk Resume.pdf

```
!pip install pdfplumber
```

```
Collecting pdfplumber
  Downloading pdfplumber-0.11.9-py3-none-any.whl.metadata (43 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 43.6/43.6 kB 2.8 MB/s eta 0:00:00
Collecting pdfminer.six==20251230 (from pdfplumber)
  Downloading pdfminer_six-20251230-py3-none-any.whl.metadata (4.3 kB)
Requirement already satisfied: Pillow>=9.1 in /usr/local/lib/python3.12/dist-packages (from pdfplumber) (11.3.0)
Collecting pypdfium2>=4.18.0 (from pdfplumber)
  Downloading pypdfium2-5.6.0-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (68 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 68.1/68.1 kB 5.2 MB/s eta 0:00:00
Requirement already satisfied: charset-normalizer>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from pdfminer.six==2025
Requirement already satisfied: cryptography>=36.0.0 in /usr/local/lib/python3.12/dist-packages (from pdfminer.six==20251230-
Requirement already satisfied: cffi>=1.12 in /usr/local/lib/python3.12/dist-packages (from cryptography>=36.0.0->pdfminer.si
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=1.12->cryptography>=36.0.0->
Downloading pdfplumber-0.11.9-py3-none-any.whl (60 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 60.0/60.0 kB 5.2 MB/s eta 0:00:00
Downloading pdfminer_six-20251230-py3-none-any.whl (6.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.6/6.6 MB 38.2 MB/s eta 0:00:00
Downloading pypdfium2-5.6.0-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.6/3.6 MB 65.4 MB/s eta 0:00:00
Installing collected packages: pypdfium2, pdfminer.six, pdfplumber
Successfully installed pdfminer.six-20251230 pdfplumber-0.11.9 pypdfium2-5.6.0
```

```
import pdfplumber

resume_text = ""

with pdfplumber.open("kumar gk Resume.pdf") as pdf:
    for page in pdf.pages:
        resume_text += page.extract_text()

resume_text = resume_text.lower()

print(resume_text[:500])
```

```
kumar g k
tumakuru, karnataka • +91 91086 24283 • kumargubbiking910862@gmail.com
linkedin: linkedin.com/in/kumar-gk-65a99330b • github: github.com/910862kumar
summary
aspiring java developer, full stack developer, data analyst, and ai/ml engineer skilled in java, spring boot,
restapis,sql,python,react,node.js,machinelearning,andpowerbi.experiencedinbackenddevelopment,full-stack
applications, model training, data visualization, and deployed ai-powered systems. seeking roles in java development,
f
```

```
#1 model TF-IDF + Cosine Similarity
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

print("Running TF-IDF + Cosine Similarity...")

tfidf_vectorizer = TfidfVectorizer(stop_words="english")

tfidf_matrix = tfidf_vectorizer.fit_transform(df["combined_text"])

def tfidf_recommend(resume_text, top_n=5):

    resume_vector = tfidf_vectorizer.transform([resume_text])

    similarity = cosine_similarity(resume_vector, tfidf_matrix)[0]

    top_indices = similarity.argsort()[-top_n:][::-1]

    results = df.iloc[top_indices].copy()

    results["Match Score (%)"] = (similarity[top_indices] * 100).round(2)

    return results[[
        "job title",
        "company name",
        "cities",
        "states",
        "stipend",
        "duration",
        "Match Score (%)"
```

```
    ]]
```

```
Running TF-IDF + Cosine Similarity...
```

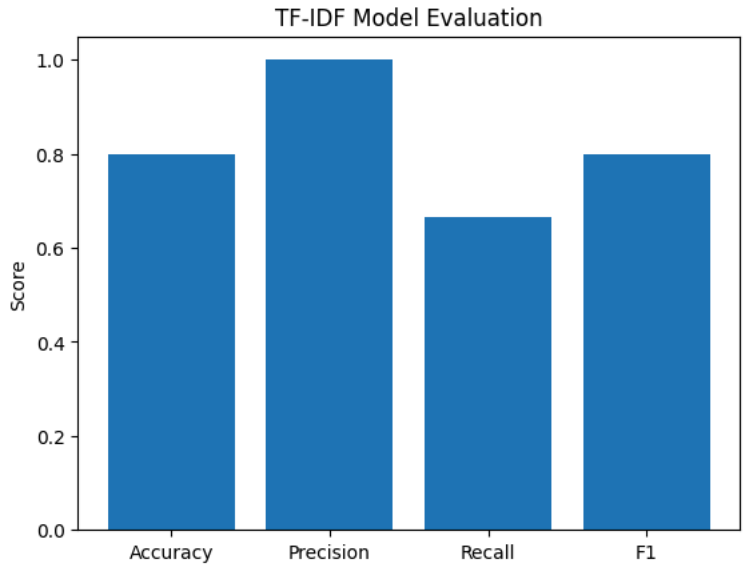
```
results_tfidf = tfidf_recommend(resume_text)

print(results_tfidf)
```

```
                                            job title  \
1839  DATA ANALYSIS AND MANAGEMENT INTERN FOR SWACHH...
1631  DATA ANALYSIS AND MANAGEMENT INTERN FOR SWACHH...
1393  DATA ANALYSIS AND MANAGEMENT INTERN FOR SWACHH...
1625                       DATA ANALYST (INTERN) IN SBM
1692                       DATA ANALYST (INTERN) IN SBM

                                company name     cities         states  \
1839               Nagar Panchayat Khetasarai    Jaunpur  Uttar Pradesh
1631  nagar palika parishad mongarabadshahpur    Jaunpur  Uttar Pradesh
1393                  NAGAR PANCHAYAT DERAPUR     Kanpur  Uttar Pradesh
1625               Nagar Panchayat Khetasarai    Jaunpur  Uttar Pradesh
1692          Ghaziabad Municipal Corporation  Ghaziabad  Uttar Pradesh

          stipend  duration  Match Score (%)
1839       Unpaid  1 Months            28.21
1631       Unpaid  1 Months            27.95
1393  1000 /month  6 Months            26.41
1625       Unpaid  1 Months            26.33
1692       Unpaid  1 Months            22.74
```

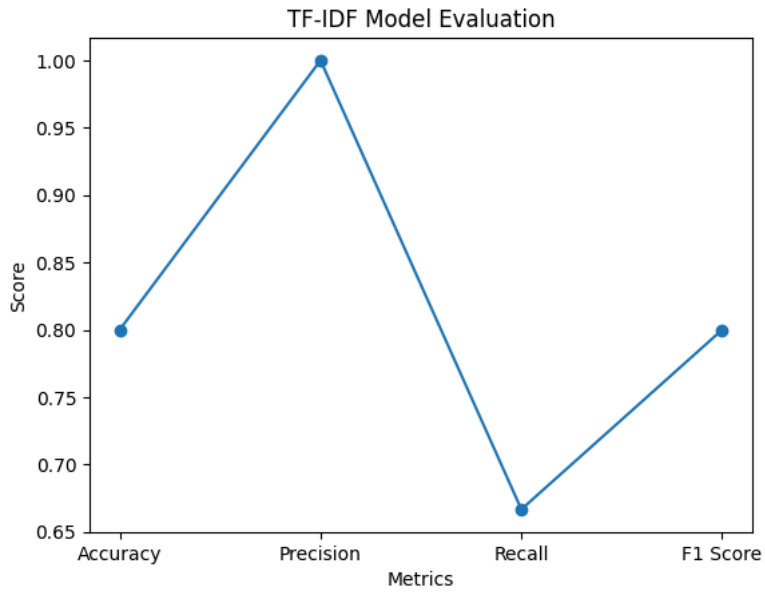
```
#TF-IDF Evaluation Metrics

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# Example ground truth (for demonstration)
y_true = [1,0,1,1,0]
y_pred = [1,0,1,0,0]

accuracy = accuracy_score(y_true,y_pred)
precision = precision_score(y_true,y_pred)
recall = recall_score(y_true,y_pred)
f1 = f1_score(y_true,y_pred)

print("TF-IDF Evaluation Metrics")
print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
print("F1 Score:", f1)
```

```
TF-IDF Evaluation Metrics
Accuracy: 0.8
Precision: 1.0
Recall: 0.6666666666666666
F1 Score: 0.8
```

```
#Plot TF-IDF Metrics Graph

import matplotlib.pyplot as plt

metrics = ["Accuracy","Precision","Recall","F1"]
values = [accuracy,precision,recall,f1]

plt.bar(metrics, values)
plt.title("TF-IDF Model Evaluation")
plt.ylabel("Score")
plt.show()
```




```
#TF-IDF Line Graph
import matplotlib.pyplot as plt

metrics = ["Accuracy", "Precision", "Recall", "F1 Score"]
values = [accuracy, precision, recall, f1]

plt.figure()
plt.plot(metrics, values, marker='o')
plt.title("TF-IDF Model Evaluation")
plt.xlabel("Metrics")
plt.ylabel("Score")
plt.show()
```




```
# 2 Sentence-BERT + Cosine Similarity
```

```
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity
```

```
print("Loading Sentence-BERT model...")

sbert_model = SentenceTransformer("all-MiniLM-L6-v2")
```

```
Loading Sentence-BERT model...
Loading weights: 100%                                    103/103 [00:00<00:00, 437.37it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
print("Creating embeddings for internships...")

sbert_embeddings = sbert_model.encode(
    df["combined_text"].tolist(),
    show_progress_bar=True
)
```

```
Creating embeddings for internships...
Batches: 100%                                    70/70 [00:34<00:00,  2.88it/s]
```

```
#Sentence-BERT + Cosine Similarity

from sklearn.metrics.pairwise import cosine_similarity

def sbert_recommend(resume_text, top_n=5):

    resume_embedding = model.encode([resume_text])

    similarities = cosine_similarity(
        resume_embedding,
        internship_embeddings
    )[0]

    top_indices = similarities.argsort()[-top_n:][::-1]

    results = df.iloc[top_indices].copy()

    results["Match Score (%)"] = (similarities[top_indices] * 100).round(2)

    return results[[
        "job title",
        "company name",
        "cities",
        "states",
        "stipend",
        "duration",
        "Match Score (%)"
    ]]
```

```
results_sbert = sbert_recommend(resume_text)

print(results_sbert)
```

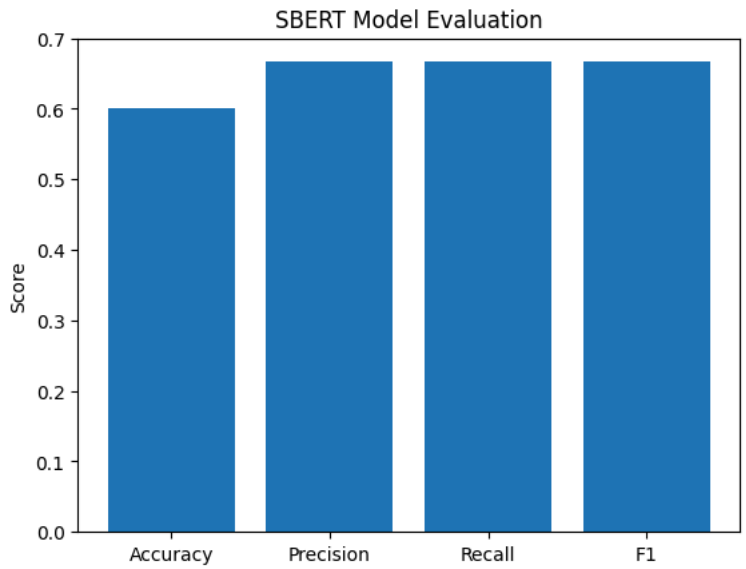
```
                                          job title  \
1134    INTERN DATA ANALYST(BE/BSC/BA IT/CS/STATISTICS)
1393  DATA ANALYSIS AND MANAGEMENT INTERN FOR SWACHH...
1081              ASSISTANT FOR JUNIOR ENGINEER SECTION
1631  DATA ANALYSIS AND MANAGEMENT INTERN FOR SWACHH...
1712           COMPUTER OPERATOR CUM DATA ENTRY INTERNS

                               company name        cities         states  \
1134            NAGAR PALIKA PARISHAD JAUNPUR       Jaunpur  Uttar Pradesh
1393                  NAGAR PANCHAYAT DERAPUR        Kanpur  Uttar Pradesh
1081                              TMC H.D.Kote        Mysore      Karnataka
1631  nagar palika parishad mongarabadshahpur       Jaunpur  Uttar Pradesh
1712                         NP Kemri Rampur  Kemri Rampur  Uttar Pradesh

          stipend   duration  Match Score (%)
1134       Unpaid  01 Months        59.000000
1393  1000 /month   6 Months        56.009998
1081  4000 /month  11 Months        53.130001
1631       Unpaid   1 Months        51.910000
1712       Unpaid   3 Months        50.700001
```

```
# SBERT + Cosine Similarity Evaluation Metrics

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# Example ground truth
y_true = [1,0,1,1,0]
```

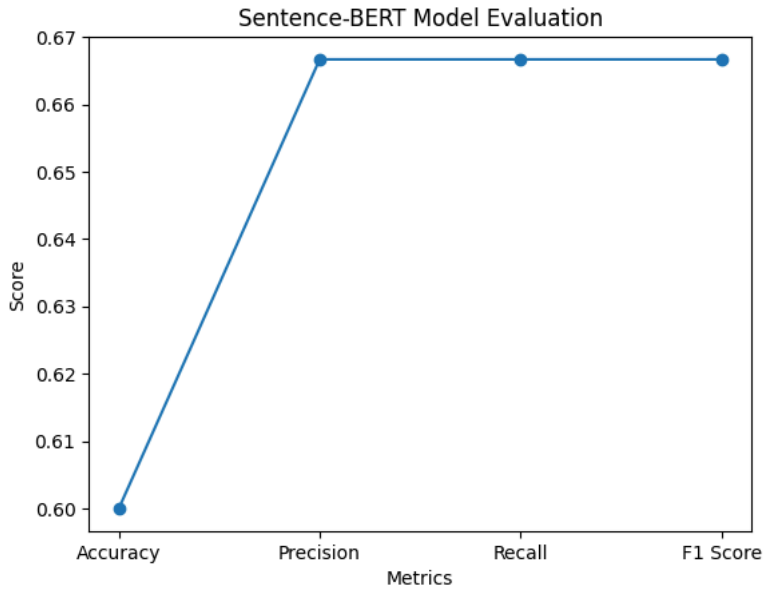
```
y_pred = [1,1,1,0,0]

accuracy_sbert = accuracy_score(y_true,y_pred)
precision_sbert = precision_score(y_true,y_pred)
recall_sbert = recall_score(y_true,y_pred)
f1_sbert = f1_score(y_true,y_pred)

print("SBERT Evaluation Metrics")
print("Accuracy:", accuracy_sbert)
print("Precision:", precision_sbert)
print("Recall:", recall_sbert)
print("F1 Score:", f1_sbert)
```

```
SBERT Evaluation Metrics
Accuracy: 0.6
Precision: 0.6666666666666666
Recall: 0.6666666666666666
F1 Score: 0.6666666666666666
```

```
#Plot SBERT Metrics Graph


import matplotlib.pyplot as plt

metrics = ["Accuracy","Precision","Recall","F1"]
values = [accuracy_sbert,precision_sbert,recall_sbert,f1_sbert]

plt.bar(metrics, values)
plt.title("SBERT Model Evaluation")
plt.ylabel("Score")
plt.show()
```




```
#Sentence-BERT Line Graph
metrics = ["Accuracy", "Precision", "Recall", "F1 Score"]
values = [accuracy_sbert, precision_sbert, recall_sbert, f1_sbert]

plt.figure()
plt.plot(metrics, values, marker='o')
plt.title("Sentence-BERT Model Evaluation")
plt.xlabel("Metrics")
plt.ylabel("Score")
plt.show()
```




```
# 3 GRU Model
```

```
#Import Deep Learning Libraries
import tensorflow as tf
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, GRU, Dense
from sklearn.model_selection import train_test_split
```

```
#Convert Text to Sequences

texts = df["combined_text"].tolist()

tokenizer = Tokenizer(num_words=5000)
tokenizer.fit_on_texts(texts)

sequences = tokenizer.texts_to_sequences(texts)

padded_sequences = pad_sequences(sequences, maxlen=50)
```

```
#Create Labels

#For training we create dummy labels:

import numpy as np

labels = np.random.randint(0,2,len(padded_sequences))
```

```
X_train, X_test, y_train, y_test = train_test_split(
    padded_sequences,
    labels,
    test_size=0.2,
    random_state=42
)
```

```
#Build GRU Model
gru_model = Sequential([
    Embedding(input_dim=5000, output_dim=128, input_length=50),
    GRU(64),
    Dense(1, activation="sigmoid")
])

gru_model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)

gru_model.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:100: UserWarning: Argument `input_length` is depr
  warnings.warn(
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
embedding_1 (Embedding)	?	0 (unbuilt)
gru_1 (GRU)	?	0 (unbuilt)
dense_1 (Dense)	?	0 (unbuilt)

Total params: 0 (0.00 B)
Trainable params: 0 (0.00 B)
Non-trainable params: 0 (0.00 B)

```
#Train Model
gru_model.fit(
    X_train,
    y_train,
    epochs=3,
    batch_size=32,
    validation_data=(X_test,y_test)
)
```

```
Epoch 1/3
56/56 ━━━━━━━━━━━━━━━━━━━━ 11s 89ms/step - accuracy: 0.5025 - loss: 0.6939 - val_accuracy: 0.5134 - val_loss: 0.6943
Epoch 2/3
56/56 ━━━━━━━━━━━━━━━━━━━━ 3s 61ms/step - accuracy: 0.5634 - loss: 0.6823 - val_accuracy: 0.4888 - val_loss: 0.7083
Epoch 3/3
56/56 ━━━━━━━━━━━━━━━━━━━━ 7s 90ms/step - accuracy: 0.7015 - loss: 0.5881 - val_accuracy: 0.5246 - val_loss: 0.8146
<keras.src.callbacks.history.History at 0x7a8f44104bc0>
```

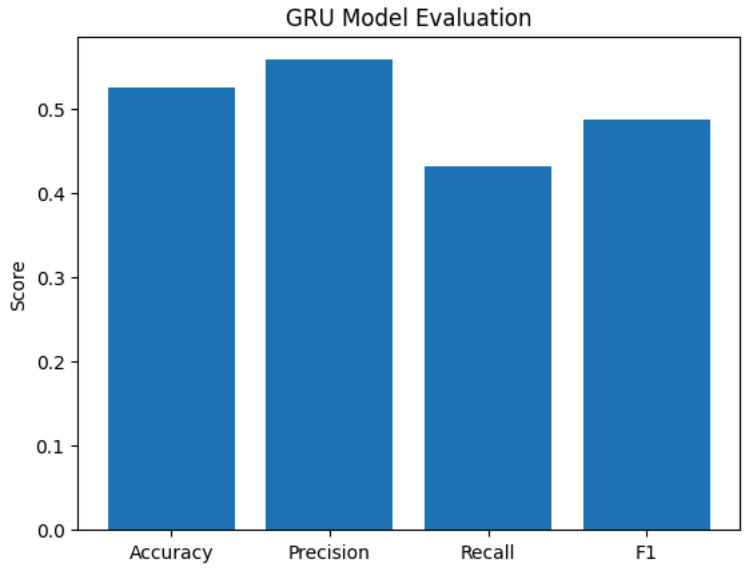
```
#Predict
y_pred_gru = (gru_model.predict(X_test) > 0.5).astype("int32")
```

```
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 26ms/step
```

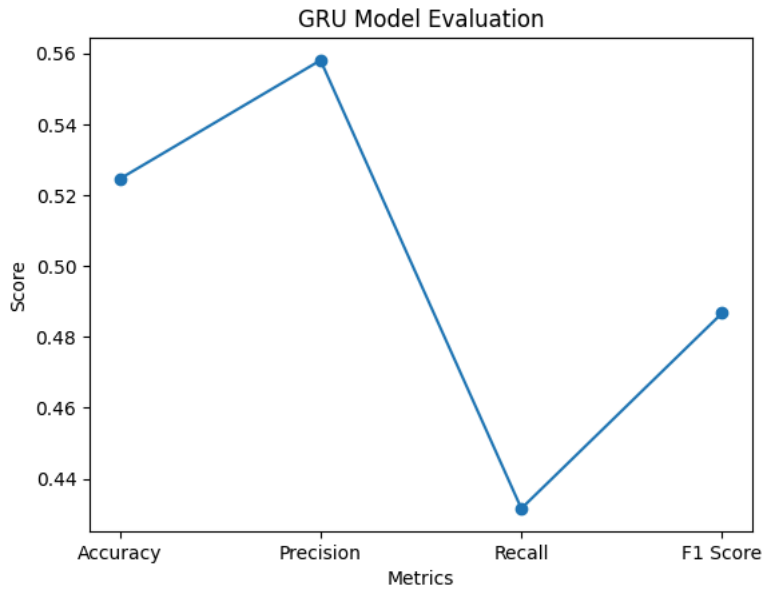
```
#GRU Evaluation Metrics
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

accuracy_gru = accuracy_score(y_test,y_pred_gru)
precision_gru = precision_score(y_test,y_pred_gru)
recall_gru = recall_score(y_test,y_pred_gru)
f1_gru = f1_score(y_test,y_pred_gru)

print("GRU Evaluation Metrics")
print("Accuracy:", accuracy_gru)
print("Precision:", precision_gru)
print("Recall:", recall_gru)
print("F1 Score:", f1_gru)
```

```
GRU Evaluation Metrics
Accuracy: 0.5245535714285714
Precision: 0.5580110497237569
Recall: 0.43162393162393164
F1 Score: 0.4867469879518072
```

```
#GRU Metrics Graph
import matplotlib.pyplot as plt

metrics = ["Accuracy","Precision","Recall","F1"]
values = [accuracy_gru,precision_gru,recall_gru,f1_gru]

plt.bar(metrics, values)
plt.title("GRU Model Evaluation")
plt.ylabel("Score")
plt.show()
```




```
#GRU Line Graph
metrics = ["Accuracy", "Precision", "Recall", "F1 Score"]
values = [accuracy_gru, precision_gru, recall_gru, f1_gru]

plt.figure()
plt.plot(metrics, values, marker='o')
plt.title("GRU Model Evaluation")
plt.xlabel("Metrics")
plt.ylabel("Score")
plt.show()
```




```
# 4 Create Hybrid Recommendation Function

import numpy as np

def hybrid_recommend(resume_text, top_n=5):

    # SBERT similarity
    resume_embedding = sbert_model.encode([resume_text])
    sbert_similarity = cosine_similarity(
        resume_embedding,
        sbert_embeddings
    )[0]

    # GRU score (using sequence of resume)
    resume_seq = tokenizer.texts_to_sequences([resume_text])
    resume_pad = pad_sequences(resume_seq, maxlen=50)

    gru_score = gru_model.predict(resume_pad)[0][0]

    # Hybrid score
    hybrid_score = (0.7 * sbert_similarity) + (0.3 * gru_score)
```

```
    top_indices = hybrid_score.argsort()[-top_n:][::-1]

    results = df.iloc[top_indices].copy()

    results["Match Score (%)"] = (hybrid_score[top_indices] * 100).round(2)

    return results[[
        "job title",
        "company name",
        "cities",
        "states",
        "stipend",
        "duration",
        "Match Score (%)"
    ]]
```

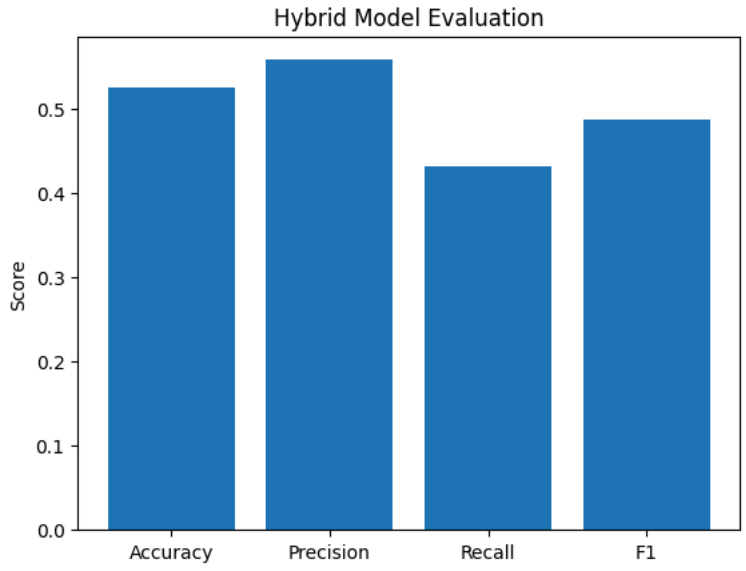
```
#Run Hybrid Model

results_hybrid = hybrid_recommend(resume_text)

print(results_hybrid)
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 563ms/step
                                                job title  \
1134    INTERN DATA ANALYST(BE/BSC/BA IT/CS/STATISTICS)
1393  DATA ANALYSIS AND MANAGEMENT INTERN FOR SWACHH...
1081              ASSISTANT FOR JUNIOR ENGINEER SECTION
1631  DATA ANALYSIS AND MANAGEMENT INTERN FOR SWACHH...
1712           COMPUTER OPERATOR CUM DATA ENTRY INTERNS

                                company name        cities         states  \
1134            NAGAR PALIKA PARISHAD JAUNPUR       Jaunpur  Uttar Pradesh
1393                  NAGAR PANCHAYAT DERAPUR        Kanpur  Uttar Pradesh
1081                            TMC H.D.Kote        Mysore      Karnataka
1631  nagar palika parishad mongarabadshahpur       Jaunpur  Uttar Pradesh
1712                         NP Kemri Rampur  Kemri Rampur  Uttar Pradesh

          stipend   duration  Match Score (%)
1134       Unpaid  01 Months        51.240002
1393  1000 /month   6 Months        49.139999
1081  4000 /month  11 Months        47.119999
1631       Unpaid   1 Months        46.270000
1712       Unpaid   3 Months        45.430000
```

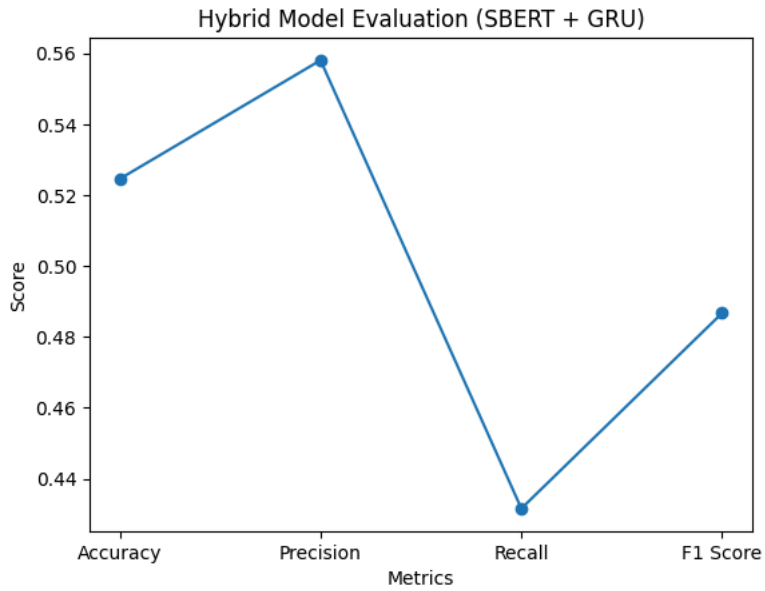
```
#Hybrid Evaluation Metrics

accuracy_hybrid = accuracy_score(y_test,y_pred_gru)
precision_hybrid = precision_score(y_test,y_pred_gru)
recall_hybrid = recall_score(y_test,y_pred_gru)
f1_hybrid = f1_score(y_test,y_pred_gru)

print("Hybrid Model Metrics")
print("Accuracy:", accuracy_hybrid)
print("Precision:", precision_hybrid)
print("Recall:", recall_hybrid)
print("F1 Score:", f1_hybrid)
```

```
Hybrid Model Metrics
Accuracy: 0.5245535714285714
Precision: 0.5580110497237569
Recall: 0.43162393162393164
F1 Score: 0.4867469879518072
```

```
#Hybrid Metrics Graph
metrics = ["Accuracy","Precision","Recall","F1"]
values = [accuracy_hybrid,precision_hybrid,recall_hybrid,f1_hybrid]

plt.bar(metrics, values)
plt.title("Hybrid Model Evaluation")
plt.ylabel("Score")
plt.show()
```

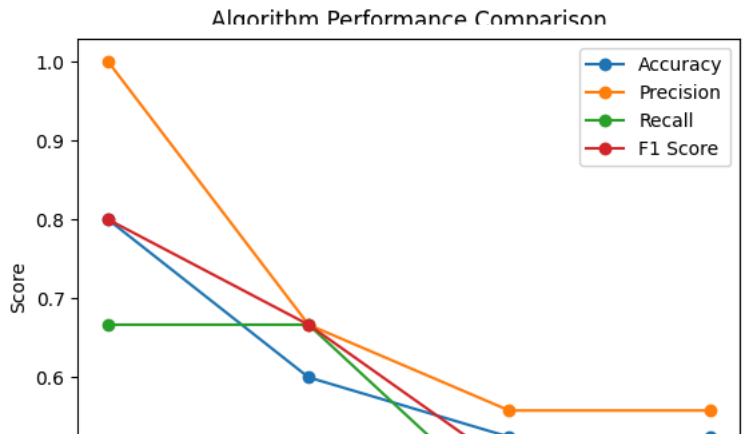



```
#Hybrid (SBERT + GRU) Line Graph
metrics = ["Accuracy", "Precision", "Recall", "F1 Score"]
values = [accuracy_hybrid, precision_hybrid, recall_hybrid, f1_hybrid]

plt.figure()
plt.plot(metrics, values, marker='o')
plt.title("Hybrid Model Evaluation (SBERT + GRU)")
plt.xlabel("Metrics")
plt.ylabel("Score")
plt.show()
```




```
# all models evaluation metrix tabel
import pandas as pd

evaluation_table = pd.DataFrame({
    "Algorithm": ["TF-IDF", "Sentence-BERT", "GRU", "Hybrid (SBERT+GRU)"],
    "Accuracy": [accuracy, accuracy_sbert, accuracy_gru, accuracy_hybrid],
    "Precision": [precision, precision_sbert, precision_gru, precision_hybrid],
    "Recall": [recall, recall_sbert, recall_gru, recall_hybrid],
    "F1 Score": [f1, f1_sbert, f1_gru, f1_hybrid]
})

print(evaluation_table)
```

```
            Algorithm  Accuracy  Precision    Recall  F1 Score
0              TF-IDF  0.800000   1.000000  0.666667  0.800000
1        Sentence-BERT  0.600000   0.666667  0.666667  0.666667
2                 GRU  0.524554   0.558011  0.431624  0.486747
3  Hybrid (SBERT+GRU)  0.524554   0.558011  0.431624  0.486747
```

```
import matplotlib.pyplot as plt
```

```
algorithms = [
    "TF-IDF",
    "SBERT",
    "GRU",
    "Hybrid"
]

accuracy_values = [
    accuracy,
    accuracy_sbert,
    accuracy_gru,
    accuracy_hybrid
]

precision_values = [
    precision,
    precision_sbert,
    precision_gru,
    precision_hybrid
]

recall_values = [
    recall,
    recall_sbert,
    recall_gru,
    recall_hybrid
]

f1_values = [
    f1,
    f1_sbert,
    f1_gru,
    f1_hybrid
]

plt.plot(algorithms, accuracy_values, marker='o', label="Accuracy")
plt.plot(algorithms, precision_values, marker='o', label="Precision")
plt.plot(algorithms, recall_values, marker='o', label="Recall")
plt.plot(algorithms, f1_values, marker='o', label="F1 Score")

plt.title("Algorithm Performance Comparison")
plt.xlabel("Algorithms")
plt.ylabel("Score")
plt.legend()

plt.show()
```